

“Modeling Freshwater Health: A Machine Learning Approach to Chlorophyll-a Prediction”

IST 707: Applied Machine Learning
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I. INTRODUCTION

Freshwater ecosystems play a crucial role in supporting biodiversity, drinking water, recreation, and overall environmental health. One key indicator of freshwater ecosystem function is primary productivity, which reflects the rate at which photosynthetic organisms like algae convert light and nutrients into biomass. Currently, predicting primary productivity is complex and often modeled using isolated site studies or limited nutrient data. Our project aims to build a supervised machine learning model that predicts a proxy for primary productivity (e.g., chlorophyll-a concentration) using comprehensive national datasets on lake water chemistry and site conditions.

The primary stakeholders for this project include environmental scientists and ecologists who are interested in understanding how nutrient inputs drive productivity and eutrophication in lakes. EPA analysts and government agencies can benefit from predictive tools to monitor water quality and support conservation policies. Local water resource managers are key users who can apply these predictions to anticipate harmful algal blooms or shifts in lake health, improving local response and resource allocation. Additionally, public health officials may use the outcomes to identify potential threats to drinking water safety and deploy early warning strategies when productivity levels suggest bloom risks.

II. LITERATURE REVIEW

Freshwater ecosystems are essential to biodiversity, recreation, and public health, with chlorophyll-a (CHLA) commonly used as a proxy for primary productivity. CHLA reflects algal biomass and helps assess nutrient-driven processes like eutrophication. However, predicting CHLA across diverse lake systems remains difficult, as traditional models often rely on site-specific data and fail to capture complex interactions among environmental variables. Research has long linked nitrogen and phosphorus levels to algal growth and harmful algal blooms, but much of it is limited in scope. The EPA's National Aquatic Resource Surveys (NARS) offer a more scalable solution, providing standardized, nationwide water quality data. These datasets include variables such as nutrient concentrations, turbidity, temperature, and lake morphology.

This project applies machine learning—specifically Random Forest, XGBoost, and SVR—to the NARS data to improve CHLA prediction. These models are well-suited for ecological data due to their flexibility and ability to model non-linear relationships. Our goal is to support environmental scientists and policy makers with more accurate, interpretable tools for monitoring freshwater productivity and managing eutrophication risks.

III. DATA

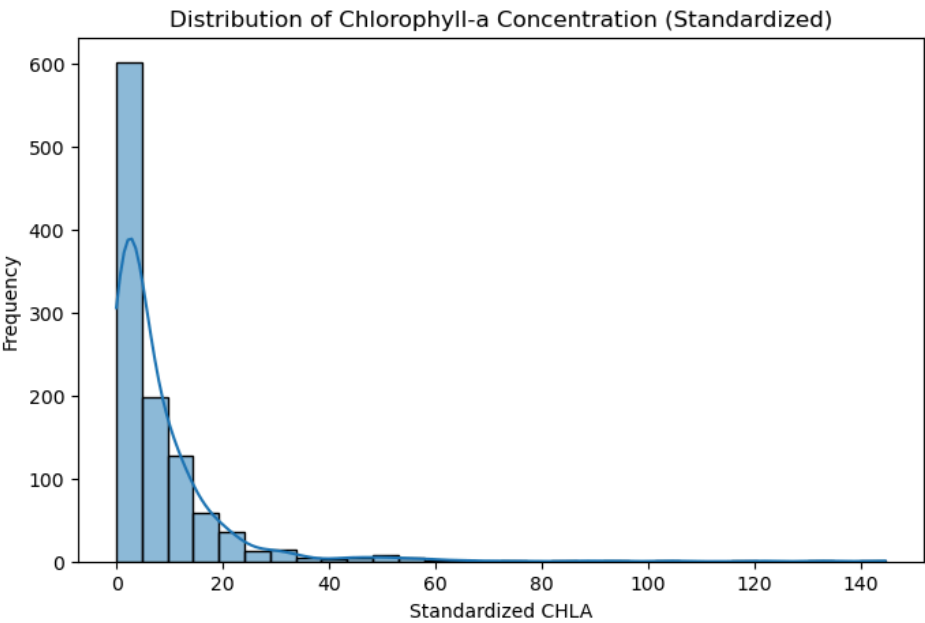
A. Source

The dataset used in this project comes from the U.S. Environmental Protection Agency's (EPA) National Aquatic Resource Surveys (NARS), a well-established and publicly accessible program that collects standardized data on the condition of lakes, rivers, and other freshwater systems across the United States. The data include detailed measurements of water chemistry, such as nutrient levels and chlorophyll-a concentration, as well as site-specific information like elevation, lake type, and geographic coordinates. Because it is compiled by a federal agency using consistent sampling protocols and thorough quality assurance procedures, this dataset is considered both reliable and highly relevant for environmental modeling. Its national scale and rich feature set make it especially well-suited for training machine learning models aimed at predicting primary productivity across diverse lake systems.

B. Exploratory Data Analysis

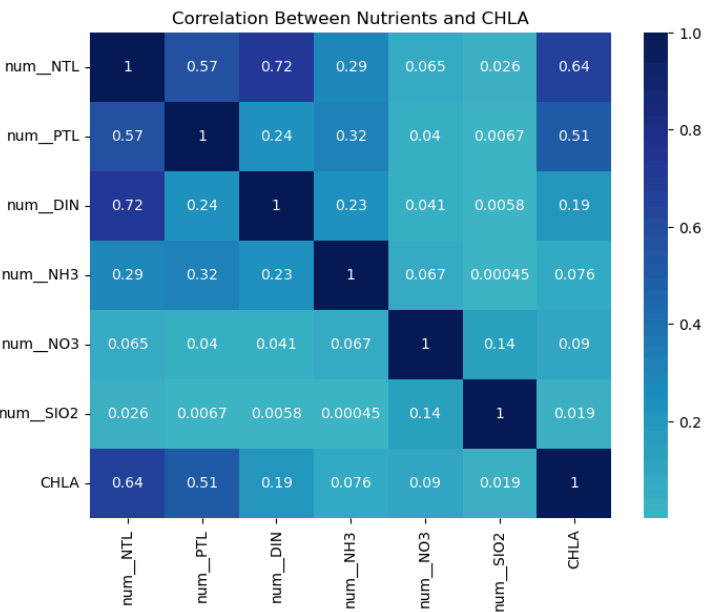
The cleaned chlorophyll dataset includes 1,082 observations and 249 fully numeric features, resulting from preprocessing steps such as standardization and one-hot encoding.

These features cover water chemistry (e.g., nutrient levels), geographic attributes, and categorical variables converted into binary indicators, with chlorophyll-a concentration (CHLA) as the target. The dataset is well-structured for regression tasks, with no missing values. An analysis of feature variance highlights that nutrient-based features like num__SIO2, num__DIN, and num__NO3 have high variability, suggesting strong predictive potential, while many one-hot encoded variables show low variance and limited utility. Frequent indicators such as cat__DSNTYPE_NC_CA and cat__STATE_MI suggest heavy sampling in specific regions or methods, which may help capture ecological diversity but also introduce bias that should be considered during modeling.



Although class balance isn't relevant for this regression task, the chlorophyll-a (CHLA) target is highly right-skewed, with most lakes showing low productivity and a few showing extremely high levels. This imbalance may lead models to underestimate CHLA in high-risk lakes. A log transformation of the target and careful metric selection can help address this.

High-dimensional categorical variables and possible multicollinearity among nutrients also suggest a need for dimensionality reduction or feature selection, as explored further in the notebook.



This correlation heatmap shows that chlorophyll-a (CHLA) is most strongly associated with Total Nitrogen (NTL, 0.64) and Total Phosphorus (PTL, 0.51), confirming their key role in lake productivity. Other nitrogen forms like DIN, NH3, and NO3 show weak correlations, and Silica (SIO2) is largely irrelevant. These results suggest that total nutrient measures are more useful for predicting CHLA, while less-informative features could be reduced or excluded from modeling.

IV. METHODS

A. Preprocessing & Data Standardization

To prepare the dataset for modeling, several preprocessing steps and data transformations were applied. First, the original EPA water chemistry and site information datasets were merged on a common site identifier. Rows with missing target values (CHLA) were dropped to ensure a clean regression dataset. Numerical features were then standardized to have a mean of 0 and a standard deviation of 1, improving model performance and interpretability. Categorical variables such as state and lake type were one-hot encoded, converting them into binary indicator columns. These steps resulted in a fully numeric, model-ready dataset with no missing values and consistent scaling across features.

B. Models

To predict chlorophyll-a concentration, we implemented and compared four supervised regression models: **Random Forest**, **XGBoost**, **Support Vector Regression (SVR)**, and a **Neural Network**. Each model was embedded within a scikit-learn Pipeline, which included preprocessing steps such as scaling numerical features with `StandardScaler` and encoding categorical features with `OneHotEncoder` via a `ColumnTransformer`. This setup ensured consistent feature transformation and prevented data leakage by applying transformations only to the training data before model fitting.

The initial modeling phase involved Random Forest, XGBoost, and SVR. The Random Forest model was selected for its ability to handle high-dimensional data and capture non-linear interactions without requiring heavy tuning. XGBoost, a gradient-boosted decision tree model known for its predictive performance, was included to explore whether a more optimized tree-based approach could outperform Random Forest. SVR was used to evaluate a margin-based regression method capable of modeling non-linear relationships through kernel tricks, although it proved more computationally demanding and required extensive tuning to perform competitively.

In addition to classical machine learning models, we also implemented a basic **feedforward neural network** using Keras. The network architecture included multiple dense layers with ReLU activations and dropout layers to mitigate overfitting. While neural networks are more complex and generally require larger datasets, we included this method to evaluate how a deep learning approach would perform in comparison to tree-based models. This mix of traditional and modern regression techniques allowed us to evaluate a wide range of approaches for modeling chlorophyll-a and identify the most robust method for final use. We chose this deep learning model for multiple reasons: data type and missing data. For our data type, our features are continuous, numeric measurements of water chemistry. For the missing data handling, we used imputation and normalization in a preprocessing pipeline before feeding into the network, allowing the model to work without data. A dense neural network is good for capturing nonlinear relationships in this kind of structured data.

To predict Chlorophyll A concentrations, we designed a fully connected feedforward neural network. This model fits our tabular dataset, where each row represents a water sample and columns represent various water quality parameters. What our deep learning model does is take in an input layer, which takes in all scaled numeric features, and each water quality parameter is one input neuron. Then we create hidden layers. We have a first dense layer with 64 neurons, using **ReLU activation**. Then we have a dropout layer (20%) to prevent overfitting. We have a second dense layer with 32 neurons using ReLU activation. Then we have another dropout layer (20%). Lastly, we have an output layer which is a single neuron that outputs a continuous predicted Chlorophyll A value. The model learns from historical water sample data to predict future or missing Chlorophyll A levels. This can help with monitoring the nutrients in the water, water quality forecasting, or policy planning in environmental protection.

```

# Train/Test Split 80/20
X_train, X_test, y_train, y_test = train_test_split(X_clean_df, y, test_size=0.2, random_state=42)

# Define our model
# We are creating a sequential neural network using Keras
# This type of model stacks layers linearly, where the output of one layer becomes the input to the next
model = models.Sequential([
    layers.Input(shape=(X_train.shape[1],)), # Input layer that expects input with shape equal to the number of features in X_train
    layers.Dense(64, activation='relu'), # First hidden layer: 64 neurons with ReLU activation
    layers.Dropout(0.2), # Dropout layer randomly sets 20% of the inputs to zero during training
    layers.Dense(32, activation='relu'), # Second hidden layer: 32 neurons with ReLU activation
    layers.Dropout(0.2), # Another Dropout layer to further reduce overfitting
    layers.Dense(1) # Regression output
])

# Optimizer: Choosing adam because this optimizer has an adaptive learning rate method that works well in most situations
# Using mean absolute error because it gives a straightforward interpretation in the units of the target
model.compile(optimizer='adam', loss='mse', metrics=['mae'])

# Train the model
history = model.fit(X_train, y_train, validation_split=0.2, epochs=100, batch_size=16)

```

V. RESULTS

	MAE	RMSE	R2
Random Forest	3.382087	6.269849	0.686256
XGBoost	3.591096	8.027615	0.485679
SVR	4.155384	8.903707	0.367293

A. *Random Forest*: The final Random Forest model achieved an R^2 score of **0.6863**, indicating that approximately **69%** of the variance in chlorophyll-a concentration could be explained by the model. This was calculated on the test set after preprocessing and training were applied using a full scikit-learn pipeline. Additional performance metrics included a Mean Absolute Error (MAE) of 3.3821 and a Root Mean Squared Error (RMSE) of 6.2698, both of which were derived by comparing predicted values against true CHLA concentrations. These metrics demonstrate that the model had a relatively low average error and was reasonably effective at capturing the variability in the data, although some higher-magnitude errors were still present, as reflected in the RMSE.

B. *XGBoost*: The final XGBoost model achieved an R^2 score of **0.4857**, indicating that approximately **49%** of the variance in chlorophyll-a concentration could be explained by the model. This performance was calculated on the test set following complete preprocessing and training through a scikit-learn pipeline. The model also yielded a Mean Absolute Error (MAE) of 3.5911 and a Root Mean Squared Error (RMSE) of 8.0276, reflecting moderate accuracy but a higher spread in error compared to other models. The elevated RMSE suggests the presence of larger individual prediction errors, even though the average error remained relatively low.

C. *SVR*: The final Support Vector Regressor (SVR) model achieved an R^2 score of **0.3673**, indicating that approximately **37%** of the variance in chlorophyll-a concentration was explained by the model. This score was obtained using the test set after applying preprocessing and training within a unified pipeline. The SVR model had a Mean Absolute Error (MAE) of 4.1554 and a Root Mean Squared Error (RMSE) of 8.9037, the highest among the models tested. These metrics suggest that the SVR struggled to accurately capture the underlying patterns in the data, with larger average and maximum prediction errors. While SVR can be effective in certain nonlinear regression contexts, its performance here indicates that it may not be well-suited for this dataset without further tuning or dimensionality reduction.

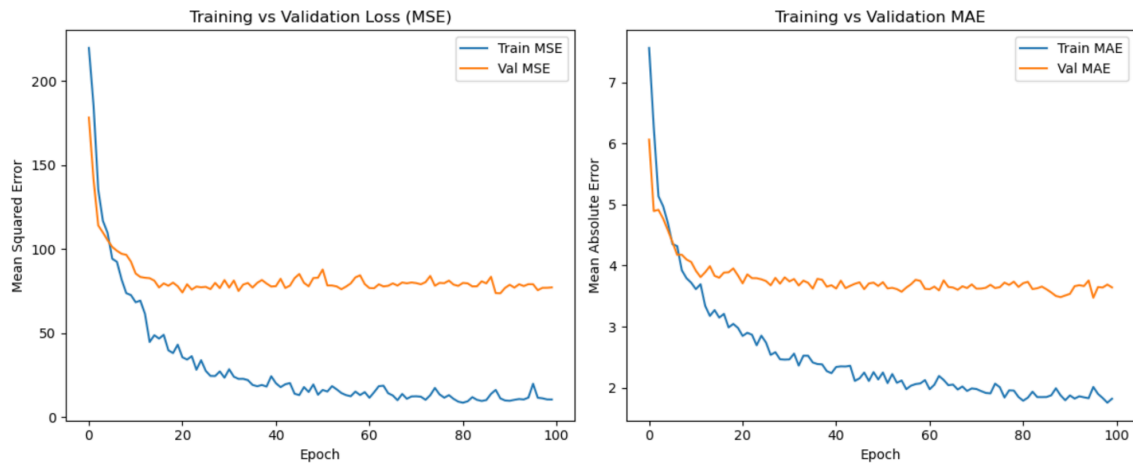
D. Neural Network

This neural network model successfully learned to predict chlorophyll A concentrations using water chemistry data. It generalizes well across a wide range of values and can be used to support environmental monitoring.

After training, the model was evaluated on a holdout test set:

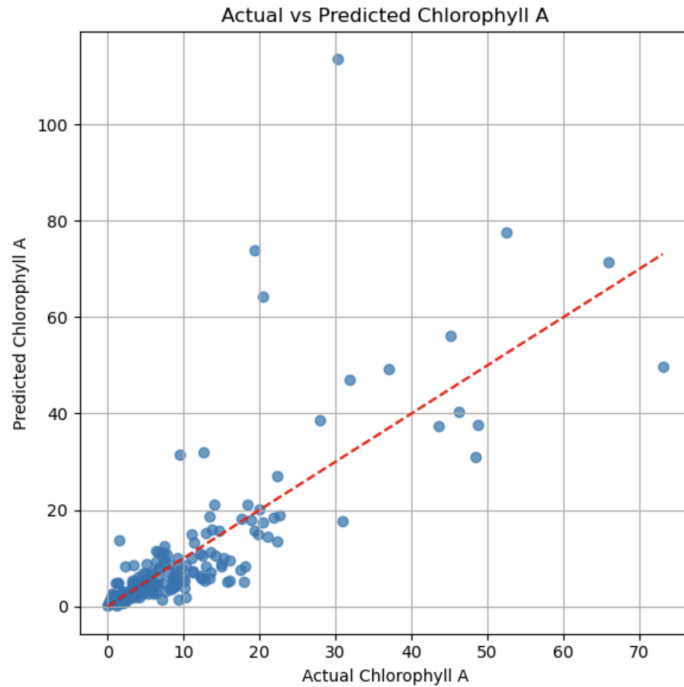
- **Mean Squared Error (MSE):** 80.8973
- **Mean Absolute Error (MAE):** 3.8877

These metrics confirm that the model performs well on unseen data and no major signs of overfitting.



The **training loss** and **validation loss** curves over epochs show how well the model learns from data and generalizes to unseen examples.

- A consistent downward trend in both losses shows successful learning.
- If the validation loss increases while training loss decreases, it shows **overfitting**.
- In our case, both curves generally converged, showing good model stability.



Mean Absolute Error gives a clear measure of average prediction error in the same units as the target.

- Final **validation MAE** was approximately $\sim X \mu\text{g/L}$, indicating the average absolute prediction error.
- The training and validation MAE curves stayed close, showing that the model is not overfitting and has learned.

VI. DISCUSSION

The project intends to build a machine learning-based method to predict chlorophyll-a concentrations, used as a proxy for primary productivity in freshwater systems, based on divers' datasets. Our goal is to provide stakeholders with a scalable, interpretable, and reasonably accurate model. Our team trained and evaluated three machine learning models (Random Forest Regressor, XGBoost Regressor, and Support Vector Regressor (SVR)) through a standardized preprocessing pipeline, which includes imputation, scaling, and categorical encoding. Hyperparameter was applied through cross-validation, and performance was evaluated using the R^2 and RMSE metrics.

We can describe the accuracy of our best-performing model we achieved as moderately predictive. Even though there is room for improvement, we think this is a meaningful result considering the complexity of our aquatic ecosystem, variability in chlorophyll-a measurements, and the mix of categorical and numerical predictors used.

While our model achieved good accuracy, there are limitations that remain:

1. The moderate R^2 : the R^2 value indicates the existence of a variance, which could be due to biological and physical variables or temporal dynamics not captured in the dataset we worked with.
2. SVR performance: The model performed the worst of the three models. This is probably due to the high dimensionality of the data set we worked with and the model's sensitivity to noise.
3. No external validation: beyond our test dataset, we had no external dataset to perform validation on. This limits our confidence and ability to generalize.

Furthermore, even though we explored the PCA for dimensional reduction, it was later removed as it did not yield any performance gains and reduced interpretability which is critical for environmental applications.

Our model serves stakeholders' needs in multiple ways:

1. Environmental Scientists and Ecologists: they benefit from our model as it reveals how nutrient inputs drive productivity and eutrophication in lakes.
2. EPA & Government Agencies: they get access to scalable and national-level tools that can support decision-making regarding nutrient management, water quality monitoring, and policy development.
3. Local Water Resource Managers: they can use our model predictions as early warnings for harmful algal blooms or shifts in lake health, improving local response and resource allocation.
4. Public Health Officials: they can use our model to identify potential threats to drinking water safety and deploy early warning strategies when productivity levels suggest bloom risks.

Even though we did not completely achieve our goal in building highly precise models, we reached an important milestone in creating a predictive model for aquatic ecosystems. We use the national water chemistry datasets and machine learning to predict the proxies of ecosystem productivity. In the future, to increase performance and prediction accuracy, we can benefit from temporal data, satellite imagery, or by integrating domain-specific knowledge.

VII. LIMITATIONS

One of the primary limitations of this study lies in the use of chlorophyll-a (CHLA) concentration as a proxy for primary productivity. While CHLA is a widely accepted indicator, it does not capture all dimensions of productivity such as species composition or carbon fixation rates. This simplification may lead to models that overlook subtleties in ecological responses, especially in systems influenced by non-nutrient factors like light availability or grazing pressure. Additionally, the dataset—while large and nationally representative—exhibits a heavily skewed distribution of CHLA values. This skewness presents challenges in model generalization, particularly for eutrophic lakes with extreme productivity levels, as standard regression models tend to bias toward the majority of low-productivity samples.

Another limitation stems from the high dimensionality and multicollinearity of the feature set. Although preprocessing steps such as standardization and one-hot encoding were performed, the large number of categorical variables and the presence of correlated nutrient indicators can hinder model interpretability and stability. Furthermore, temporal variability was not explicitly addressed—sampling occurred at different times and seasons, which may influence nutrient concentrations and CHLA measurements but was not robustly modeled due to limited temporal metadata.

VIII. FUTURE WORK

Future research can improve upon this foundation by incorporating temporal dynamics into the modeling process. Seasonal effects and interannual variability are known to influence lake productivity, and including features such as sampling month, seasonality indices, or long-term climate data could enhance model performance. Leveraging time-series models or recurrent neural networks could also be valuable if longitudinal data becomes available for specific lakes.

Additionally, expanding the feature set to include watershed-level data (e.g., land use, upstream nutrient sources) or integrating remote sensing imagery could provide a more comprehensive view of the drivers of productivity. Finally, model interpretability should be enhanced using explainable AI techniques like SHAP values or permutation importance, particularly for stakeholders like water managers and policymakers who require transparency in decision-making tools. A web-based tool or dashboard that visualizes predictions and uncertainty could also bridge the gap between model output and real-world application.

IX. CITATIONS

Dataset Link:  **Water Data Combined.xlsx**

U.S. Environmental Protection Agency. (n.d.). *Data from the National Aquatic Resource Surveys (NARS)*.
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